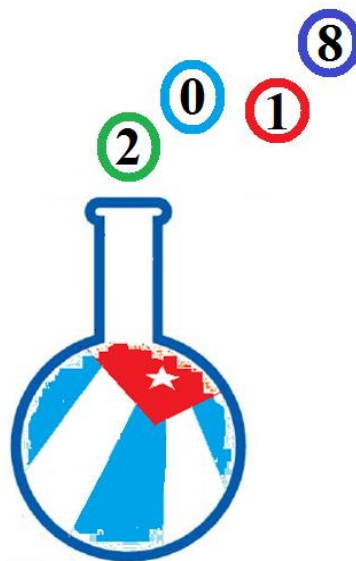


PREPARATORY PROBLEMS

SOLUTIONS



CUBA

NATIONAL CHEMISTRY

CONTEST 2018



Ministerio de Educación
República de Cuba



PREPARATORY PROBLEMS

SOLUTIONES

Cuba

National Chemistry Contest 2018

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Free distribution

Preface

To the olympists

The present material has been created for the Cuban students and mentors participating in the Cuban National Chemistry Contest of 2018, corresponding to the course 2017-2018. The aim is not to substitute already available materials but to complement and, to some extent, guide the last stage of the training after the Provincial Contest.

These Preparatory Problems were carefully selected to exercise the topics of particular difficulties, or *Advanced topics*, that are explicitly declared. These topics are the most problematic contents within the syllabus (see Annex 3) that will be evaluated in the next edition of the olympiad. This does not mean that other program topics will be excluded from the exam. I would like to let you know that the degree of difficulty and the extension of the problems are, in the majority of cases, superior to the questions in the next national exam, which corresponds with the pedagogical objective of this booklet. The answers will be explained during the National Seminar for Mentor Teachers held in Havana next October.

The booklet also includes annexes with the results of the Cuban participants in the most recent national and international events. In the future, we'll be working on elaborating an updated contest program and developing the Preparatory Problems with the contribution of mentor teachers from all over the country.

Acknowledgements

To the Ministry of Education of the Republic of Cuba to facilitate the distribution of this material. To the Faculty of Chemistry for supporting the training and laboratory practice. Likewise, I congratulate national and provincial mentors for contributing to the scientific education of thousands of students.

The author

Havana, September 2017

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CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Cero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Water ionic product at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																		
1 1A												13 3A		14 4A	15 5A	16 6A	17 7A	18 8A
1 H 1.008	2 2A											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18	
3 Li 6.941	4 Be 9.012											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95	
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80	
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3	
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)	
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)	
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)							

Proposed Tasks

Advanced topics

Common exam

Kinetics: Analysis of complex reaction mechanism by the Steady State approximation for intermediates.

Stoichiometry: Calculations involving transformation relations, sample compositions and empirical formulae of inorganic compounds.

Inorganic: Chemical properties of the representative and transition elements, including the identification of ions/substances.

10th grade

Thermochemistry: Calculations of ΔH using thermodynamic cycles.

General: Representation of molecules and description of covalent bonding by Lewis Formula, MO theory, VSREPR model, resonance theory and hybridization.

11th grade

Equilibrium: Relation between the thermodynamic magnitudes (ΔH , ΔS , ΔG , ΔE) and the equilibrium constant; applications in the gaseous phase equilibrium and in the redox process; Latimer diagrams of standard electrode potentials.

Ionic equilibrium: Calculation of PH in complex systems. Acid-base titrations, titration curve and acid-base indicators.

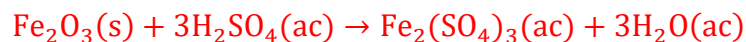
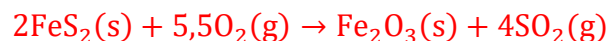
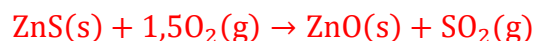
12th grade

Equilibrium: relation between the thermodynamic magnitudes (ΔH , ΔS , ΔG , ΔE) and the equilibrium constant; applications in redox processes; Latimer diagrams of standard electrode potentials

Organic: Chemical reactions that involve aldehyde and ketones.

Problem 1. A sulfide mixture

1.1.



1.2.

$$n(\text{SO}_2) = \frac{p \cdot V(\text{SO}_2)}{R \cdot T} = \frac{101,325 \cdot 1,90}{8,3145 \cdot 298,15} = 0,0777 \text{ mol}$$

$$n(\text{SO}_2) = 8n(\text{S}_8) + n(\text{ZnS}) + 2n(\text{FeS}_2)$$

$$8n(\text{S}_8) + n(\text{ZnS}) + 2n(\text{FeS}_2) = 0,0777 \quad (1)$$

$$m(\text{ZnSO}_4) + m(\text{Fe}_2(\text{SO}_4)_3) = 4,66$$

$$n(\text{ZnSO}_4) \cdot M(\text{ZnSO}_4) + n(\text{Fe}_2(\text{SO}_4)_3) \cdot M(\text{Fe}_2(\text{SO}_4)_3) = 4,66$$

$$n(\text{ZnS}) \cdot M(\text{ZnSO}_4) + 1/2 \cdot n(\text{FeS}_2) \cdot M(\text{Fe}_2(\text{SO}_4)_3) = 4,66$$

$$161,48 \cdot n(\text{ZnS}) + 0,5 \cdot 399,91 \cdot n(\text{FeS}_2) = 4,66 \quad (2)$$

$$m(\text{S}_8) + m(\text{ZnS}) + m(\text{FeS}_2) = 4,00$$

$$256,56 \cdot n(\text{S}_8) + 97,48 \cdot n(\text{ZnS}) + 119,99 \cdot n(\text{FeS}_2) = 4,00 \quad (3)$$

Solving the system formed by equations 1, 2 and 3, one gets:

$$n(\text{S}_8) = 0,00467 \text{ mol}$$

$$n(\text{ZnS}) = 0,0102 \text{ mol}$$

$$n(\text{FeS}_2) = 0,0151 \text{ mol}$$

$$m(\text{S}_8) = 0,00467 \cdot 256,56 = 1,20 \text{ g}$$

$$\omega(\text{S}_8) = \frac{1,20}{4,00} \cdot 100 = \mathbf{30\%}$$

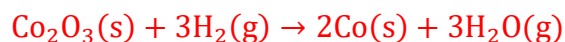
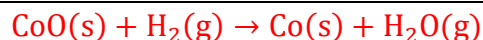
$$m(\text{ZnS}) = 0,0102 \cdot 97,48 = 0,994 \text{ g}$$

$$\omega(\text{ZnS}) = \frac{0,994}{4,00} \cdot 100 = \mathbf{25\%}$$

$$m(\text{FeS}_2) = 0,0151 \cdot 119,99 = 1,81 \text{ g}$$

$$\omega(\text{FeS}_2) = \frac{1,81}{4,00} \cdot 100 = \mathbf{45\%}$$

Problem 2. A cobalt sample



$$m(\text{Co}) + m(\text{CoO}) + m(\text{Co}_2\text{O}_3) = 5,718 \quad (1)$$

$$m(\text{Cu}) + m(\text{CoO}) + m(\text{Co}_2\text{O}_3) = 5,794 \quad (2)$$

$$m(\text{Cu}) - m(\text{Co}) = 0,076$$

$$\frac{M(\text{Cu})}{M(\text{Co})} \cdot m(\text{Co}) - m(\text{Co}) = 0,076 \quad \rightarrow \quad \frac{63,54}{58,93} \cdot m(\text{Co}) - m(\text{Co}) = 0,076$$

$$m(\text{Co}) = \frac{0,076}{\frac{63,54}{58,93} - 1} = 0,972 \text{ g}$$

$$\omega(\text{Co}) = \frac{0,972}{5,718} \cdot 100 = \mathbf{17\%}$$

$$m(\text{CoO}) + m(\text{Co}_2\text{O}_3) = 5,718 - 0,972 = 4,743 \text{ g} \quad (3)$$

$$m(\text{H}_2\text{O})_{\text{CoO}} + m(\text{H}_2\text{O})_{\text{Co}_2\text{O}_3} = 1,221$$

$$\frac{M(\text{H}_2\text{O})}{M(\text{CoO})} \cdot m(\text{CoO}) + \frac{3 \cdot M(\text{H}_2\text{O})}{M(\text{Co}_2\text{O}_3)} \cdot m(\text{Co}_2\text{O}_3) = 1,221$$

$$\frac{18,02}{74,93} \cdot m(\text{CoO}) + \frac{3 \cdot 18,02}{165,86} \cdot m(\text{Co}_2\text{O}_3) = 1,221 \quad (4)$$

Solving the system formed by equations 3 and 4, one gets

$$m(\text{CoO}) = 3,803 \text{ g} \rightarrow \omega(\text{CoO}) = \frac{3,803}{5,718} \cdot 100 = 67\%$$

$$m(\text{Co}_2\text{O}_3) = 0,940 \text{ g} \rightarrow \omega(\text{Co}_2\text{O}_3) = \frac{0,940}{5,718} \cdot 100 = 16\%$$

Problem 3. Graphite compounds I

3.1.

In compounds C_{8n}M :

$$n(\text{C}) = 8 \cdot x \cdot n(\text{M}) \rightarrow \frac{m(\text{C})}{M(\text{C})} = 8 \cdot x \cdot \frac{m(\text{M})}{M(\text{M})} \rightarrow M(\text{M}) = 8 \cdot x \cdot \frac{m(\text{M})}{m(\text{C})} \cdot M(\text{C})$$

$$M(\text{M}) = 96,08 \cdot x \cdot \frac{\omega_{\%}(\text{M})}{100 - \omega_{\%}(\text{M})}$$

E

$$M(\text{M}) = 96,08 \cdot x \cdot \frac{16,23}{100 - 16,23} = 18,61 \cdot x$$

$$x = 2 \quad M(\text{M}) = 37,22 \quad x = 3 \quad M(\text{M}) = 55,83 \quad (\text{Fe})$$



F

$$M(\text{M}) = 96,08 \cdot x \cdot \frac{24,85}{100 - 24,85} = 31,77 \cdot x$$

$$x = 2 \quad M(\text{M}) = 63,54 \quad (\text{Cu})$$



G

$$M(\text{M}) = 96,08 \cdot x \cdot \frac{22,23}{100 - 22,23} = 27,46 \cdot x$$

$$x = 2 \quad M(\text{M}) = 54,93 \quad (\text{Mn})$$



H

$$M(\text{M}) = 96,08 \cdot x \cdot \frac{23,47}{100 - 23,47} = 29,47 \cdot x$$

$$x = 2 \quad M(\text{M}) = 58,93 \quad (\text{Co})$$



In compounds $\text{MCl}_n \cdot x\text{H}_2\text{O}$

$$x = \frac{n(\text{H}_2\text{O})}{n(\text{M})} = \frac{\frac{100 - \omega_{\%}(\text{MCl}_n)}{M(\text{H}_2\text{O})}}{\frac{\omega_{\%}(\text{M})}{M(\text{M})}} \rightarrow x = \frac{M(\text{M})}{M(\text{H}_2\text{O})} \cdot \frac{100 - \frac{\omega_{\%}(\text{M})}{M(\text{M})} M(\text{MCl}_n)}{\omega_{\%}(\text{M})}$$

$$M(\text{H}_2\text{O}) = 18,02$$

A

If Fe (II), $M(\text{MCl}_n) = M(\text{FeCl}_2) = 126,75$

$$x = \frac{55,85}{18,02} \cdot \frac{100 - \frac{34,43}{55,85} \cdot 126,75}{34,43} = 1,97 \quad \text{FeCl}_2 \cdot 2\text{H}_2\text{O} ?$$

If Fe (III), $M(\text{MCl}_n) = M(\text{FeCl}_3) = 162,20$

$$x = \frac{55,85}{18,02} \cdot \frac{100 - \frac{34,43}{55,85} \cdot 162,20}{34,43} = 0,0007$$

A = FeCl₃

B

If Cu (I), $M(\text{MCl}_n) = M(\text{CuCl}) = 99,00$

$$x = \frac{63,55}{18,02} \cdot \frac{100 - \frac{37,27}{63,55} \cdot 99,00}{37,27} = 3,97 \quad \text{CuCl} \cdot 4\text{H}_2\text{O} ?$$

If Cu (II), $M(\text{MCl}_n) = M(\text{CuCl}_2) = 134,45$

$$x = \frac{55,85}{18,02} \cdot \frac{100 - \frac{34,43}{55,85} \cdot 134,45}{55,85} = 2,001$$

B = CuCl₂ · 2H₂O

C

If Mn (II), $M(\text{MCl}_n) = M(\text{MnCl}_2) = 125,84$

$$x = \frac{54,94}{18,02} \cdot \frac{100 - \frac{27,76}{54,94} \cdot 125,84}{27,76} = 3,999$$

C = MnCl₂ · 4H₂O

D

If Co (II), $M(\text{CoCl}_n) = M(\text{CoCl}_2) = 129,83$

$$x = \frac{58,93}{18,02} \cdot \frac{100 - \frac{24,77}{58,93} \cdot 129,83}{24,77} = 5,998$$



3.2.

In I (100g)

$$n(\text{C}) = \frac{\omega_{\%}(\text{C})}{M(\text{C})} = \frac{46,35}{12,01} = 3,859$$

$$n(\text{Fe}) = \frac{3}{35} \cdot 3,859 = 0,3308 \quad \rightarrow \quad m(\text{Fe}) = 0,3308 \cdot 55,85 = 18,47$$

$$n(\text{Cl}) = \frac{100 - 46,35 - 18,47}{35,45} = 0,9924$$

$$n(\text{C}) : n(\text{Cl}) : n(\text{Fe}) = 3,859 : 0,9924 : 0,3308 = 11,67 : 3 : 1 = 70 : 18 : 6$$



Oxidation numbers: C +1/70, Cl -1, Fe +2 y +3

In J (100g)

$$n(\text{C}) = \frac{\omega_{\%}(\text{C})}{M(\text{C})} = \frac{42,68}{12,01} = 3,554$$

$$n(\text{Fe}) = \frac{1}{9} \cdot 3,554 = 0,3949 \quad \rightarrow \quad m(\text{Al}) = 0,3949 \cdot 26,98 = 10,65$$

$$n(\text{Cl}) = \frac{100 - 42,68 - 10,65}{35,45} = 1,3165$$

$$n(\text{C}) : n(\text{Cl}) : n(\text{Al}) = 3,554 : 1,3165 : 0,3949 = 9 : 3,334 : 1 = 27 : 10 : 3$$



Oxidation numbers: C +1/27, Cl -1, Al +3

Problem 4. Graphite compounds II

4.1.

In **L** (100g)

$$n(\text{P}) = \frac{24,59}{30,97} = 0,7940 \text{ mol}$$

$$n(\text{F}) = \frac{100 - 24,59}{19,00} = 3,969 \text{ mol}$$

$$n(\text{F}):n(\text{P}) = 3,969:0,7940 = 5,00:1$$



In **M** (100g)

$$n(\text{Si}) = \frac{26,99}{28,09} = 0,9608 \text{ mol}$$

$$n(\text{F}) = \frac{100 - 26,99}{19,00} = 3,843 \text{ mol}$$

$$n(\text{F}):n(\text{P}) = 3,843:0,9608 = 4,00:1$$



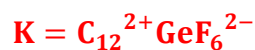
In **K** (100g)

According to the chemical equation, the ration C : Ge is constant, only the fluorine content changes. Accordingly,

$$\frac{\omega_{\%}(\text{C})}{100} = \frac{12 \cdot 12,01}{12 \cdot 12,01 + 72,64 + x \cdot M(\text{F})}$$

$$x \cdot M(\text{F}) = \frac{12 \cdot 12,01 \cdot 100}{43,58} - 12 \cdot 12,01 - 72,64 = 113,94$$

$$x = \frac{113,94}{19,00} = 5,997 \approx 6$$



In **N** (100g)

N should contain phosphorous. Supposing that the preparation reaction is of exchange (non-redox) and because compound $\text{C}_{24}^+\text{SiF}_5^-$ can be conveniently represented as $\text{C}_{24}^+\text{F}^- \cdot \text{SiF}_4$, the reaction will be better represented as:



$$M(\text{PF}_5) = 30,97 + 5 \cdot 19,00 = 125,97 \text{ g mol}^{-1}$$

$$\frac{\omega_{\%}(\text{C})}{100} = \frac{24 \cdot 12,01}{24 \cdot 12,01 + 19,00 + x \cdot M(\text{PF}_5)}$$

$$x \cdot M(\text{PF}_5) = \frac{24 \cdot 12,01 \cdot 100}{51,55} - 24 \cdot 12,01 - 19,00 = 251,91$$

$$x = \frac{251,91}{125,97} = 1,997 \approx 2$$



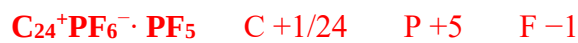
In **O** (100g)

As **N** loses PF_5 , **O** should be $\text{C}_{24}^{+}\text{PF}_6^{-}$. Checking,

$$\frac{\omega_{\%}(\text{C})}{100} = \frac{24 \cdot 12,01}{24 \cdot 12,01 + 30,97 + 6 \cdot 19,00} = 0,6654 \text{ which matches the data}$$

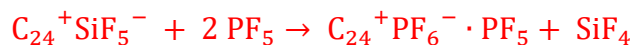


Formal oxidation numbers:



4.2.

Balanced chemical equations



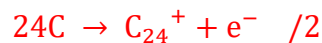
4.3.

In **P** (100g)

$$n(\text{C}) = \frac{49,57}{12,01} = 4,127 \quad n(\text{O}) = \frac{33,02}{16,00} = 2,064 \quad n(\text{S}) = \frac{16,54}{32,07} = 0,5157$$

$$n(\text{H}) = \frac{100 - 49,57 - 33,02 - 16,54}{1,008} = 0,8631$$

$$n(\text{C}) : n(\text{O}) : n(\text{H}) : n(\text{S}) = 8,00 : 4,00 : 1,67 : 1 \approx 24 : 12 : 5 : 3$$



In **Q** (100g)

Because its structure is similar to **P**, its formula must be $\mathbf{C}_{24}^+ \mathbf{NO}_3^- \cdot x\mathbf{HNO}_3$

$$\mathbf{M}(\mathbf{HNO}_3) = 1,008 + 14,01 + 3 \cdot 16,00 = 63,02 \text{ gmol}^{-1} \quad \mathbf{M}(\mathbf{NO}_3^-) = 62,01 \text{ gmol}^{-1}$$

$$\frac{\omega_{\%}(\mathbf{C})}{100} = \frac{24 \cdot 12,01}{24 \cdot 12,01 + 62,01 + x \cdot \mathbf{M}(\mathbf{HNO}_3)}$$

$$x \cdot \mathbf{M}(\mathbf{HNO}_3) = \frac{24 \cdot 12,01 \cdot 100}{60,52} - 24 \cdot 12,01 - 62,01 = 126,02$$

$$x = \frac{126,02}{63,02} = 1,9997 \approx 2$$



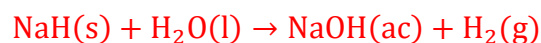
Problem 5. Sodium compounds

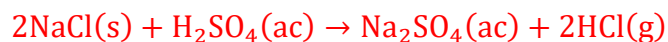
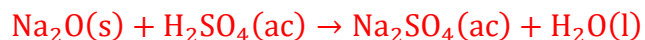
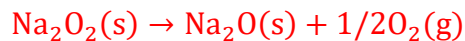
5.1.



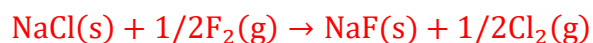
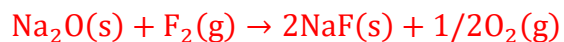
Y = $\mathbf{F}_2$ and **E** = $\mathbf{NaF}$ are accepted.

5.2.



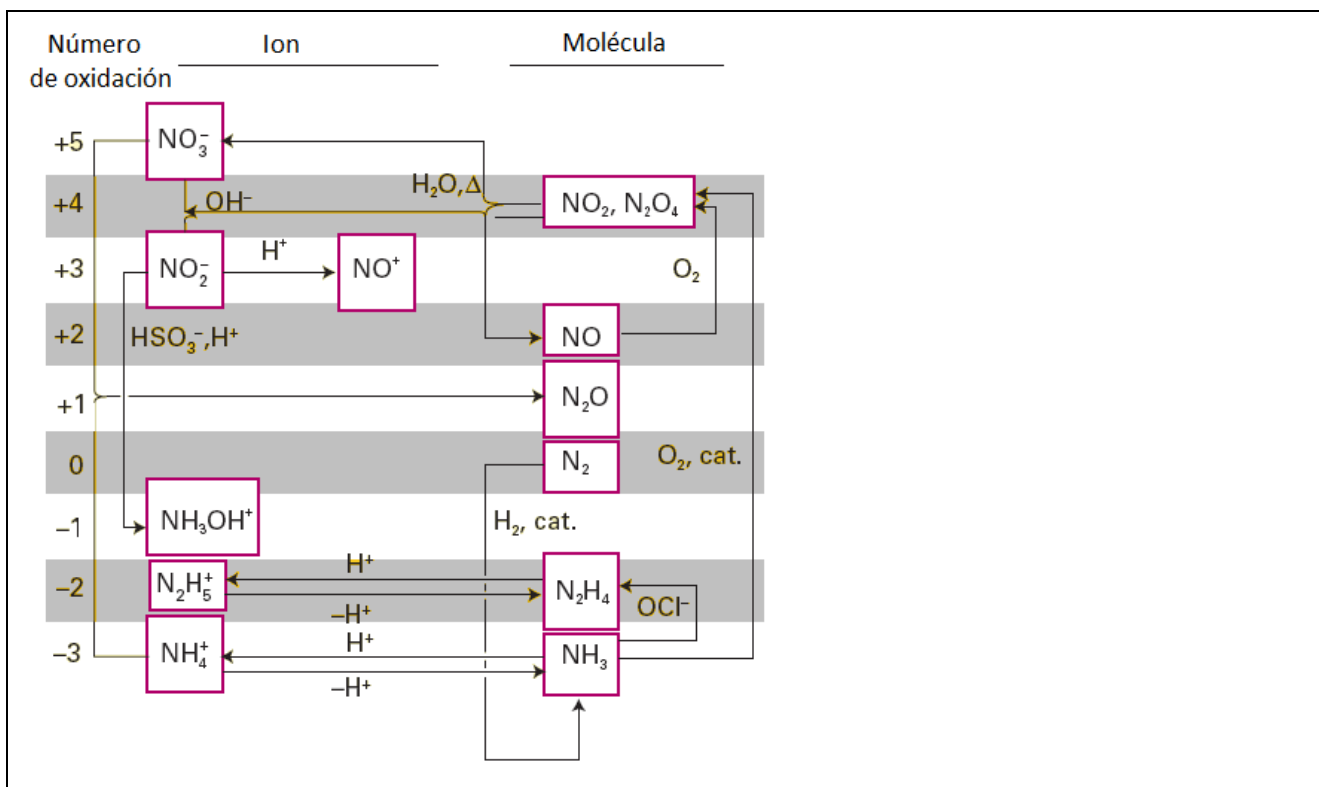


If $\text{Y} = \text{F}_2$ and $\text{E} = \text{NaF}$, then:



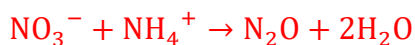
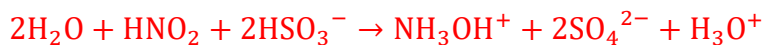
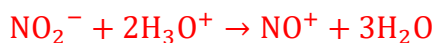
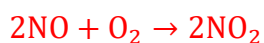
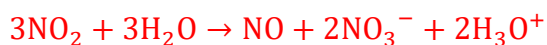
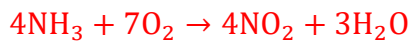
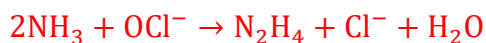
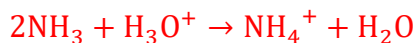
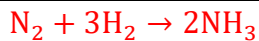
Problem 6. Nitrogen compounds

6.1. ¹

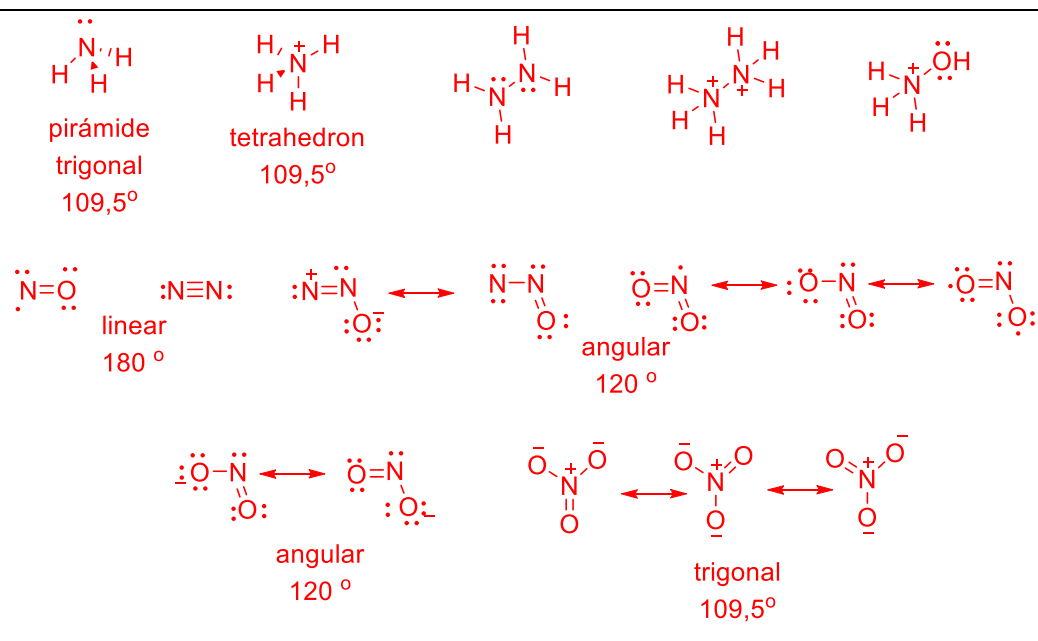


¹ Atkins, P.; Overton, T.; Rourke, J.; Weller, M.; Armstrong, F.; Hagerman, M., The Group 15 elements. In Shriver and Atkins' Inorganic Chemistry; Fifth Edition; W. H. Freeman and Company; New York, United States of America 2010, 387.

6.2.



6.4.



Problem 7. Chemical bonding

7.1.

If **A** = X_2O_a

If **B** = X_2O_b

$$\omega(X/A) = \frac{2 \cdot A_r(X)}{2 \cdot A_r(X) + aA_r(O)}$$

$$\omega(X/B) = \frac{2 \cdot A_r(X)}{2 \cdot A_r(X) + bA_r(O)}$$

$$\frac{\omega(X/A)}{\omega(X/B)} = \frac{\frac{2 \cdot A_r(X)}{2 \cdot A_r(X) + aA_r(O)}}{\frac{2 \cdot A_r(X)}{2 \cdot A_r(X) + bA_r(O)}} = \frac{2 \cdot A_r(X) + bA_r(O)}{2 \cdot A_r(X) + aA_r(O)} = \frac{2 \cdot A_r(X) + 16 \cdot b}{2 \cdot A_r(X) + 16 \cdot a} = 1,25$$

$$2 \cdot A_r(X) + 16 \cdot b = 1,25 \cdot (2 \cdot A_r(X) + 16 \cdot a)$$

$$16 \cdot (b - 1,25 \cdot a) = 0,5 \cdot A_r(X) \rightarrow A_r(X) = 32 \cdot (b - 1,25 \cdot a)$$

By try and failure one gets $a = 4$, $b = 6$ and $A_r(X) = 32$. Then $X = S$.

Checking this result in sulfur oxoacids

In H_2SO_3

$$\omega\left(\frac{S}{H_2SO_3}\right) = \frac{A_r(S)}{A_r(S) + 2 \cdot A_r(H) + 3 \cdot A_r(O)} = \frac{32,07}{32,07 + 2 \cdot 1,01 + 3 \cdot 16,00} = 39,1\%$$

In H_2SO_4

$$\omega\left(\frac{S}{H_2SO_4}\right) = \frac{A_r(S)}{A_r(S) + 2 \cdot A_r(H) + 4 \cdot A_r(O)} = \frac{32,07}{32,07 + 2 \cdot 1,01 + 4 \cdot 16,00} = 32,7\%$$

Another way of solution can be $b - a = 2$ taking into account that the most usual oxidation numbers of the elements change in two units, generally.

$$\frac{2 \cdot A_r(X) + 16 \cdot b}{2 \cdot A_r(X) + 16 \cdot a} = 1,25 \rightarrow \frac{2 \cdot A_r(X) + 16 \cdot (a + 2)}{2 \cdot A_r(X) + 16 \cdot a} = 1,25$$

Para $a = 4$, $A_r(X) = 32$ y $X = S$.

7.2.

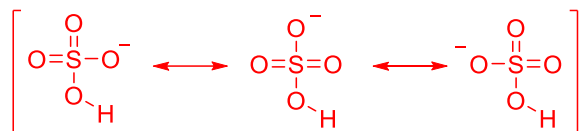
A = SO_2

B = SO_3

C = H_2SO_3

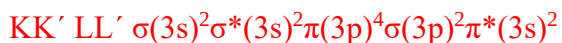
D = H_2SO_4

7.3.



7.4.

The electronic structure of S₂ molecule is



The energetic diagram instead of the electronic distribution is accepted.

$$\text{OE} = \frac{e(\text{enlaz}) - e(\text{ant})}{2} = \frac{8 - 4}{2} = 2$$

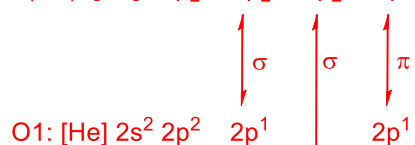
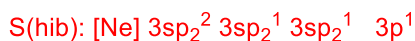
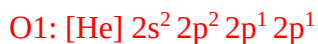
As the bond order is different from 0, the S₂ molecules should be stable.

There are unpaired electron therefore the molecule must be paramagnetic .

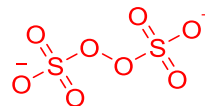
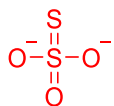
1

7.5.

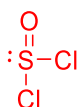
A possible representation can be



7.6.



7.7.



trigonal pyramid

~120°

Problem 8. Identification of inorganic substances I

8.1.

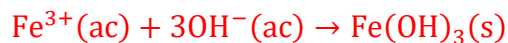
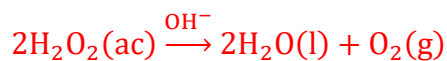
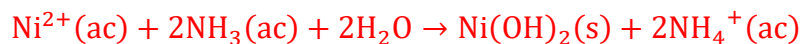
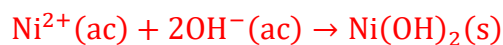
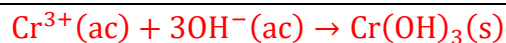
A = CrCl₃ **B** = NiCl₂ **C** = H₂O₂ **D** = NaOH **E** = NH₃ **F** = FeCl₃

The grey precipitate is iron(III) hydroxide (hydrated oxide) de Fe (III), which allows to assignate flask F as FeCl₃.

When treating C with D and E gas evolution is observed, which is only possible if C is H₂O₂, which is thermodynamically unstable and its decomposition is catalyzed in basic media. Then, D and E are NH₃ and NaOH, but it is impossible to distinguish between them at this point.

The green precipitates indicate that A and B are CrCl₃ and NiCl₂. To distinguish among them the fact that chromium(III) hydroxide is amphoteric and soluble in excess hydroxide is useful, while nickel(II) hydroxide is completely basic and does not dissolve in excess. This allows to identify A as CrCl₃, B as NiCl₂, D as NaOH and E with NH₃.

8.2.



Representing [Cr(OH)₄]⁻ instead of [Cr(OH)₆]³⁻ is also correct.

8.3.

A: green

B: green

C, D y E: colorless

F = brown

Problem 9. Identification of inorganic substances II

9.1.

Mixture A: AgNO_3 , $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$, NH_4NO_3

It is colorless – there is no nickel

When treated with HCl or HNO_3 no gas evolution is observed – there is no carbonate

Only a portion is soluble in water and the white insoluble residue becomes soluble in the presence of NH_3 – AgCl precipitates

Mixture B: ZnO , $\text{Pb}(\text{NO}_3)_2$, Mg

It is grey colored – contains Mg or MnO_2

When adding water, a grey fraction does not dissolve – ZnO or BaSO_4 should be present

It dissolves slowly in HNO_3 with no solid residue and gas evolution – Mg and ZnO are present

When adding HCl , a white precipitate is observed - $\text{Pb}(\text{NO}_3)_2$ is present

KI is excluded as the yellow PbI precipitate was not observed.

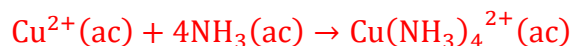
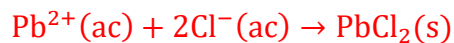
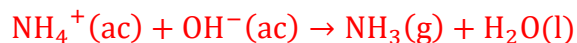
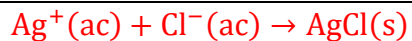
Mixture C: CaCO_3 , $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$

It dissolves completely both in HCl and in HNO_3 with gas evolution – TiO_2 is absent and CaCO_3 is present, FeSO_4 is absent because $\text{Fe}(\text{OH})_3$ is not formed

It dissolves in HCl and HNO_3 giving a blueish solution. When treated with ammonia a deep blue color is generated – there is $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$

NH_4I cannot be present together with Cu^{2+} or iodine would be formed

9.2.



Problem 10. Analysis of reaction mechanisms

10.1.

Summing up both equations from the mechanism one gets



10.2.

If R = CH₃COCH₃, reactant; I = CH₃C(OH)CH₂, intermediate; and P = CH₃COCH₂Br, product,

$$v = \frac{d(\text{P})}{dt} = k_2 c(\text{I}) c(\text{Br}_2)$$

$$\frac{d(\text{P})}{dt} = k_1 c(\text{R}) c(\text{H}^+) - k_{-1} c(\text{I}) c(\text{H}^+) - k_2 c(\text{I}) c(\text{Br}_2) = 0$$

$$c(\text{I}) = \frac{k_1 c(\text{R}) c(\text{H}^+)}{k_{-1} c(\text{H}^+) + c(\text{Br}_2)}$$

Substituting in the rate law

$$v = k_2 \frac{k_1 c(\text{R}) c(\text{H}^+)}{k_{-1} c(\text{H}^+) + c(\text{Br}_2)} c(\text{Br}_2)$$

10.3.

At high bromine concentrations

$$v = k_2 \frac{k_1 c(R) c(H^+)}{k_{-1} c(H^+) + k_2 c(Br_2)} c(Br_2) \approx \frac{k_1 c(R) c(H^+)}{c(Br_2)} c(Br_2) = k_1 c(R) c(H^+)$$

The reaction is first order with respect to CH_3COCH_3 and the ion H^+ .

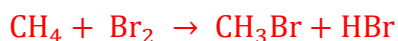
At low bromine concentrations

$$v = k_2 \frac{k_1 c(R) c(H^+)}{k_{-1} c(H^+) + k_2 c(Br_2)} c(Br_2) \approx k_2 \frac{k_1 c(R) c(H^+)}{k_{-1} c(H^+)} c(Br_2) = k' c(R) c(Br_2)$$

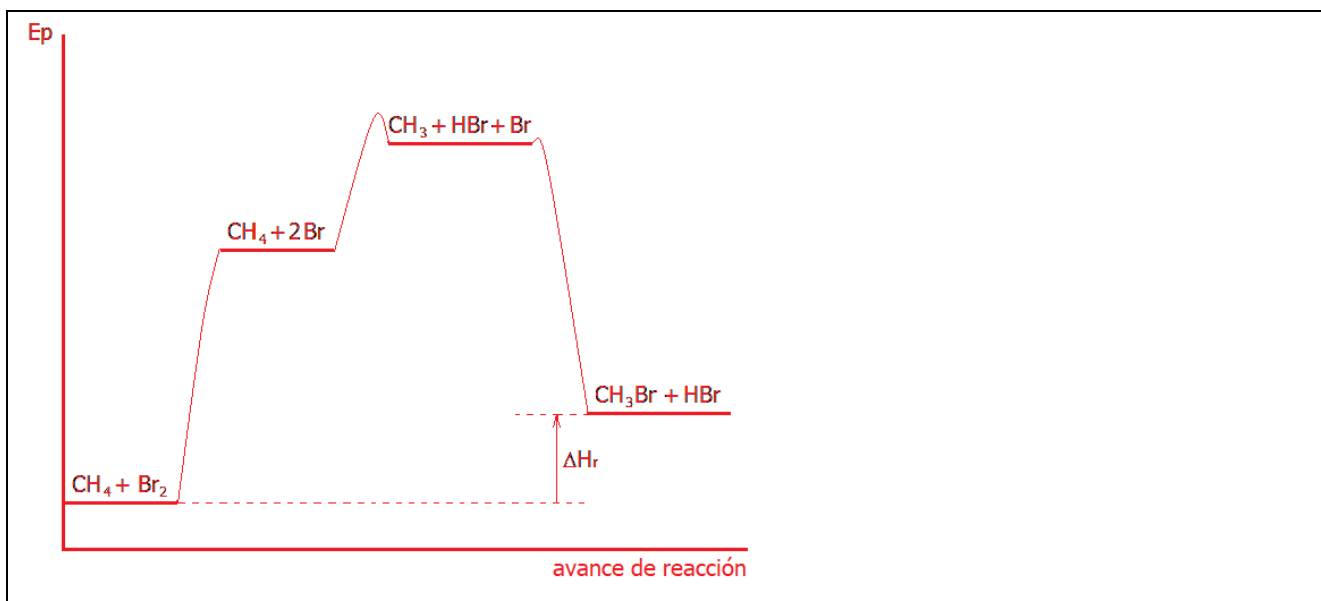
The reaction is first order with respect to CH_3COCH_3 and Br_2 .

10.4.

Summing up the second and third equations from the mechanism one gets

**10.5.**

The rate determining step is the slowest, in this case, the first one, whose activation energy is higher. It is unimolecular.

10.6.

Problem 11. Ethanal decomposition

11.1.

$$\frac{dc(\text{CH}_4)}{dt} = k_2 c(\text{CH}_3\text{CHO}) c(\text{CH}_3 \cdot)$$

$$\frac{dc(\text{CH}_3 \cdot)}{dt} = 0 = k_1 c(\text{CH}_3\text{CHO}) + k_3 c(\text{CH}_3\text{CO} \cdot) - k_2 c(\text{CH}_3\text{CHO}) c(\text{CH}_3 \cdot) - 2k_4 c^2(\text{CH}_3 \cdot)$$

$$\frac{dc(\text{CH}_3\text{CO} \cdot)}{dt} = 0 = k_2 c(\text{CH}_3\text{CHO}) c(\text{CH}_3 \cdot) - k_3 c(\text{CH}_3\text{CO} \cdot)$$

Summing up the previous equations

$$0 = k_1 c(\text{CH}_3\text{CHO}) - 2k_4 c^2(\text{CH}_3 \cdot) \rightarrow c(\text{CH}_3 \cdot) = \sqrt{\frac{k_1 c(\text{CH}_3\text{CHO})}{2k_4}}$$

$$\frac{dc(\text{CH}_4)}{dt} = k_2 \sqrt{\frac{k_1 c(\text{CH}_3\text{CHO})}{2k_4}} c(\text{CH}_3\text{CHO}) = \frac{k_2 \sqrt{k_1}}{\sqrt{2k_4}} c^{3/2}(\text{CH}_3\text{CHO})$$

11.2.

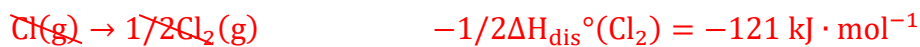
$$k = \frac{k_2 \sqrt{k_1}}{\sqrt{2k_4}}$$

11.3.

$$E_A = E_A(2) + \frac{1}{2} E_A(1) - \frac{1}{2} E_A(4)$$

Problem 12. Born-Haber cycle

12.1.





$$U(\text{NaCl}) = \Delta H_{\text{f}}^{\circ}(\text{NaCl(s)}) - \text{EA}(\text{Cl}) - 1/2\Delta H_{\text{dis}}^{\circ}(\text{Cl}_2) - I(\text{Na}) - \Delta H_{\text{sub}}^{\circ}(\text{Na})$$

$$U(\text{NaCl}) = -411 + 349 - 124 - 496 - 107 = -786 \text{ kJ} \cdot \text{mol}^{-1} \quad 1$$

12.2.

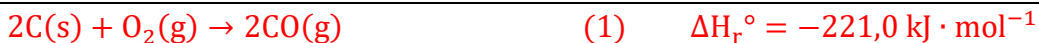
As Ag^{+} ion has a d^{10} electronic structure and is slightly larger, it is more polarizable than Na^{+} ion, in consequence, the electron density is better distributed in the lattice of AgCl and this factor increases the value of the lattice energy. There is a covalent contribution to the formation of the crystal.

12.3.

Water solubility and melting point are higher for NaCl because the ionic character of the bond in this compound is larger.

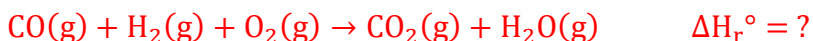
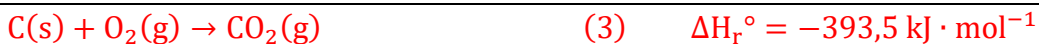
Problem 13. Coal gasification

13.1.



$$\Delta H_{\text{r}}^{\circ} = 1/2(\Delta H_{\text{r}}^{\circ}(1) - \Delta H_{\text{r}}^{\circ}(2)) = 1/2(-221,0 + 483,6) = 131,3 \text{ kJ} \cdot \text{mol}^{-1}$$

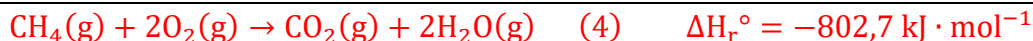
13.2.



$$\Delta H_{\text{r}}^{\circ} = \Delta H_{\text{r}}^{\circ}(3) - 1/2\Delta H_{\text{r}}^{\circ}(1) + 1/2\Delta H_{\text{r}}^{\circ}(2) = -393,5 + 1/2 \cdot 221,0 - 1/2 \cdot 483,6$$

$$\Delta H_{\text{r}}^{\circ} = -524,8 \text{ kJ} \cdot \text{mol}^{-1}$$

13.3.



$$\Delta H_{\text{r}}^{\circ} = \Delta H_{\text{r}}^{\circ}(3) + 3/2\Delta H_{\text{r}}^{\circ}(2) - 1/2\Delta H_{\text{r}}^{\circ}(1) - \Delta H_{\text{r}}^{\circ}(4) = -3 \cdot 483,6 + 1/2 \cdot 221,0 + 802,7$$

$$\Delta H_{\text{r}}^{\circ} = -393,5 - 3/2 \cdot 483,6 + 1/2 \cdot 221,0 + 802,7 = -205,7 \text{ kJ} \cdot \text{mol}^{-1}$$

Problem 14. Industrial preparation of hydrogen

14.1.

$$\Delta H_{\text{r}}^{\circ} = 3\Delta H_{\text{f}}^{\circ}(\text{H}_2) + \Delta H_{\text{f}}^{\circ}(\text{CO}) - \Delta H_{\text{f}}^{\circ}(\text{CH}_4) - \Delta H_{\text{f}}^{\circ}(\text{H}_2\text{O})$$

$$\Delta H_{\text{r}}^{\circ} = -110,5 + 74,4 + 241,8 = 205,7 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta S_{\text{r}}^{\circ} = 3S^{\circ}(\text{H}_2) + S^{\circ}(\text{CO}) - S^{\circ}(\text{CH}_4) - S^{\circ}(\text{H}_2\text{O})$$

$$\Delta S_{\text{r}}^{\circ} = 3 \cdot 130,7 + 197,7 - 186,3 - 188,8 = 214,7 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$$

$$\Delta G_{\text{r}}^{\circ} = \Delta H_{\text{r}}^{\circ} - T\Delta S_{\text{r}}^{\circ} = 205700 - 298 \cdot 214,7 = 141700 \text{ J} \cdot \text{mol}^{-1}$$

$$\Delta G_{\text{r}}^{\circ} = -TR \ln K_{\text{p}} \quad \rightarrow \quad K_{\text{p}} = \exp\left(\frac{\Delta G_{\text{r}}^{\circ}}{-TR}\right) = \exp\left(\frac{141700}{-8,314 \cdot 298}\right) = 1,44 \cdot 10^{-25}$$

14.2.

Because the reaction is endothermic, increasing the temperature shifts the equilibrium towards the formation of the products, that is, increases the equilibrium constant.

For ideal gases the volume fraction and the molar fraction are the same. Assuming that equal volumes of the reactants are mixed at the atmospheric pressure, if there is 0,2% (in volume) of CH_4 in the equilibrium mixture, the same amount of H_2O should remain. Rest 99,6% corresponds to the products H_2 and CO in ratio 3:1. Therefore there is 74,7% of H_2 and 24,9% of CO .

$$K_{\text{p}} = \frac{p_{\text{r}}^3(\text{H}_2)p_{\text{r}}(\text{CO})}{p_{\text{r}}(\text{CH}_4)p_{\text{r}}(\text{H}_2\text{O})} = \frac{\chi^3(\text{H}_2) \frac{p_{\text{Tot}}^3}{p^{\circ 3}} \cdot \chi(\text{CO}) \frac{p_{\text{Tot}}}{p^{\circ}}}{\chi(\text{CH}_4) \frac{p_{\text{Tot}}}{p^{\circ}} \cdot \chi(\text{H}_2\text{O}) \frac{p_{\text{Tot}}}{p^{\circ}}} = \frac{\chi^3(\text{H}_2) \cdot \chi(\text{CO})}{\chi(\text{CH}_4) \cdot \chi(\text{H}_2\text{O})} \frac{p_{\text{Tot}}^2}{p^{\circ 2}}$$

$$K_{\text{p}} = \frac{0,747^3 \cdot 0,249}{0,002 \cdot 0,002} \cdot \frac{101325^2}{100000^2} = 26640$$

14.3.

$$\ln\left(\frac{K_2}{K_1}\right) = -\frac{\Delta H_r^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \rightarrow$$

$$T_2 = \left(-\frac{R}{\Delta H_r^\circ} \ln\left(\frac{K_2}{K_1}\right) + \frac{1}{T_1}\right)^{-1} = \left(-\frac{8,314}{205700} \ln\left(\frac{26640}{1,44 \cdot 10^{-25}}\right) + \frac{1}{298}\right)^{-1} = 1580 \text{ K}$$

Problem 15. Hydrates**15.1.**

$$\Delta H_r^\circ = \Delta H_f^\circ(\text{H}_2\text{O}(\text{g})) - \Delta H_f^\circ(\text{H}_2\text{O}(\text{l})) = -241,8 - (-285,8) = 44,0 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta S_r^\circ = S^\circ(\text{H}_2\text{O}(\text{g})) - S^\circ(\text{H}_2\text{O}(\text{l})) = 188,7 - 70,1 = 118,6 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$$

$$\Delta G_r^\circ = \Delta H_r^\circ - T \cdot \Delta S_r^\circ = 44000 - 298 \cdot 118,6 = 8,66 \cdot 10^3 \text{ J} \cdot \text{mol}^{-1}$$

$$K^\circ = \frac{p^\circ(\text{H}_2\text{O})}{p^\circ} = \exp\left(-\frac{\Delta G_r^\circ}{R \cdot T}\right) = \exp\left(-\frac{8,66 \cdot 10^3}{8,314 \cdot 298}\right) = 0,0304$$

$$p^\circ(\text{H}_2\text{O}) = 0,0304 \cdot 100 = 3,04 \text{ kPa}$$

15.2.

$$\Delta H_r^\circ = \Delta H_f^\circ(\text{CuSO}_4 \cdot 3\text{H}_2\text{O}(\text{s})) + 2\Delta H_f^\circ(\text{H}_2\text{O}(\text{g})) - \Delta H_f^\circ(\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}))$$

$$\Delta H_r^\circ = -1683,1 - 2 \cdot 241,8 - (-2278) = 111,3 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta S_r^\circ = S^\circ(\text{CuSO}_4 \cdot 3\text{H}_2\text{O}(\text{s})) + 2S^\circ(\text{H}_2\text{O}(\text{g})) - S^\circ(\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}))$$

$$\Delta S_r^\circ = 225,1 + 2 \cdot 188,7 - 305,4 = 297,1 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$$

$$\Delta G_r^\circ = \Delta H_r^\circ - T \cdot \Delta S_r^\circ = 111300 - 298 \cdot 297,1 = 2,28 \cdot 10^4 \text{ J} \cdot \text{mol}^{-1}$$

$$K^\circ = \frac{p^2(\text{H}_2\text{O})}{p^{\circ 2}} = \exp\left(-\frac{\Delta G_r^\circ}{R \cdot T}\right) = \exp\left(-\frac{2,28 \cdot 10^4}{8,314 \cdot 298}\right) = 1,01 \cdot 10^{-4}$$

$$p(\text{H}_2\text{O}) = \sqrt{1,01 \cdot 10^{-4} \cdot 100} = 1,00 \text{ kPa}$$

15.3.

The crystal is totally hydrated, in the form of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

15.4.



At the boiling point, $p^\circ(\text{H}_2\text{O}) = p = 100 \text{ kPa}$. Then,

$$\Delta G_r^\circ = -R \cdot T \cdot \ln \frac{p(\text{H}_2\text{O})}{p^\circ} = 0 = \Delta H_r^\circ - T \cdot \Delta S_r^\circ$$

$$T = \frac{\Delta H_r^\circ}{\Delta S_r^\circ} = \frac{44000}{118,6} = 371 \text{ K} = 98 \text{ }^\circ\text{C}$$

Problem 16. Acid-base equilibrium

16.1.



$$p(\text{CO}_2) = \chi(\text{CO}_2) \cdot p = \frac{0,035}{100} \cdot 101,3 = 0,0355 \text{ kPa} = 35,5 \text{ Pa}$$

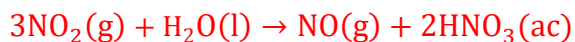
$$[\text{CO}_2] = K_H \cdot p(\text{CO}_2) = 3,36 \cdot 10^{-7} \cdot 35,5 = 1,19 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{H}^+] = \sqrt{K_{a1}(\text{CO}_2) \cdot [\text{CO}_2]}$$

$$[\text{H}^+] = \sqrt{4,47 \cdot 10^{-7} \cdot 1,19 \cdot 10^{-5}} = \sqrt{5,32 \cdot 10^{-12}} = 2,31 \cdot 10^{-6} \text{ mol} \cdot \text{L}^{-1}$$

$$\text{pH} = -\log(\text{H}^+) = -\log(2,31 \cdot 10^{-6}) = 5,64$$

16.2.



16.3.

As nitric acid is a strong acid

$$c(\text{HNO}_3) = [\text{NO}_3^-] \approx [\text{H}^+] = 10^{-\text{pH}} = 10^{-4,5} = 3,2 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

16.4.

Sulfuric acid is strong in the first dissociation step and weak in its second dissociation

$$[\text{H}^+] \approx [\text{HSO}_4^-] + 2[\text{SO}_4^{2-}] = \frac{[\text{H}^+]}{K_{a2}} \cdot [\text{SO}_4^{2-}] + 2[\text{SO}_4^{2-}] = [\text{SO}_4^{2-}] \left(\frac{[\text{H}^+]}{K_{a2}} + 2 \right)$$

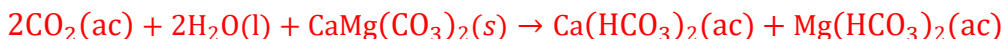
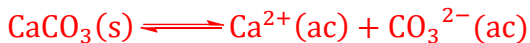
$$[\text{SO}_4^{2-}] = \frac{[\text{H}^+]}{\left(\frac{[\text{H}^+]}{K_{a2}} + 2 \right)} = \frac{3,2 \cdot 10^{-5}}{\left(\frac{3,2 \cdot 10^{-5}}{1,02 \cdot 10^{-2}} + 2 \right)} = 1,6 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{HSO}_4^-] = \frac{[\text{H}^+]}{K_{a2}} \cdot [\text{SO}_4^{2-}] = \frac{3,2 \cdot 10^{-5}}{1,02 \cdot 10^{-2}} \cdot 1,6 \cdot 10^{-5} = 5,0 \cdot 10^{-8} \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{H}_2\text{SO}_4) = [\text{HSO}_4^-] + [\text{SO}_4^{2-}] = 5,0 \cdot 10^{-8} + 1,6 \cdot 10^{-5} = 1,6 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

Alternatively, as the concentration of sulfuric acid is small and the second dissociation constant of sulfuric acid is large, the second dissociation can be considered as complete.

$$[\text{H}^+] \approx 2 \cdot c(\text{H}_2\text{SO}_4) \rightarrow c(\text{H}_2\text{SO}_4) = \frac{[\text{H}^+]}{2} = \frac{3,2 \cdot 10^{-5}}{2} = 1,6 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

16.5.**16.6.**

$$[\text{H}^+] = 10^{-\text{pH}} = 10^{-5,65} = 2,24 \cdot 10^{-6} \text{ mol} \cdot \text{L}^{-1}$$

$$\alpha_2 = \frac{[\text{CO}_3^{2-}]}{c(\text{CO}_3^{2-})} = \frac{K_{a1} \cdot K_{a2}}{[\text{H}^+]^2 + K_{a1} \cdot [\text{H}^+] + K_{a1} \cdot K_{a2}}$$

$$\alpha_2 = \frac{4,47 \cdot 10^{-7} \cdot 4,68 \cdot 10^{-11}}{(2,24 \cdot 10^{-6})^2 + 2,24 \cdot 10^{-6} \cdot 4,47 \cdot 10^{-7} + 4,47 \cdot 10^{-7} \cdot 4,68 \cdot 10^{-11}}$$

$$\alpha_2 = 3,48 \cdot 10^{-6}$$

$$\text{But } S = [\text{Ca}^{2+}] = c(\text{CO}_3^{2-})$$

$$K_{ps} = [\text{Ca}^{2+}] \cdot [\text{CO}_3^{2-}] = [\text{Ca}^{2+}] \cdot \alpha_2 \cdot c(\text{CO}_3^{2-}) = \alpha_2 \cdot S^2$$

$$S = \sqrt{\frac{K_{ps}}{\alpha_2}} = \sqrt{\frac{3,36 \cdot 10^{-9}}{3,48 \cdot 10^{-6}}} = 0,031 \text{ mol} \cdot \text{L}^{-1}$$

The calculation can proceed employing the equilibrium with dissolved CO₂.

Problem 17. Acid-base titrations

17.1.

The container must be plastic to avoid the re reaction between the base and the glass. Atmosphere must be inert because alkalis absorb CO₂ from the air. Long waiting time is needed for performing the analysis to guarantee equilibrium conditions.

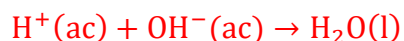
17.2.

An acid-base indicator is a weak acid or a weak base in which the molecular species and the ionic species have different colors. Hence, the resultant color of the indicator solution will depend on the pH value.

17.3.

A primary standard is a chemically stable substance, of well-defined composition, large molecular weight, inert towards the air components and that reacts quantitatively and stoichiometrically.

17.4.



$$\bar{V}(\text{HCl}) = \frac{29,4 + 29,8 + 29,6}{3} = 29,6 \text{ mL}$$

Volume measurement of 30,5 mL must be rejected.

$$c(\text{OH}^-) = \frac{n(\text{OH}^-)}{V(\text{Ca}(\text{OH})_2)} = \frac{c(\text{HCl}) \cdot \bar{V}(\text{HCl})}{V(\text{Ca}(\text{OH})_2)} = \frac{0,0150 \cdot 29,6}{20,0} = 0,0222 \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{Ca}^{2+}) = \frac{c(\text{OH}^-)}{2} = \frac{0,0222}{2} = 0,0111 \text{ mol} \cdot \text{L}^{-1}$$

$$K_{\text{ps}} = [\text{Ca}^{2+}] \cdot [\text{OH}^-]^2 = 0,0111 \cdot (0,0222)^2 = 5,47 \cdot 10^{-6}$$

17.5.

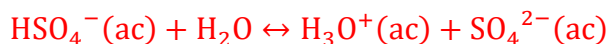


$$c(\text{H}_2\text{SO}_4)_{\text{dil}} = \frac{c(\text{NaOH}) \cdot V(\text{NaOH})}{V(\text{H}_2\text{SO}_4)_{\text{dil}} \cdot 2} = \frac{0,01000 \cdot 24,14}{10,00 \cdot 2} = 0,01207 \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{H}_2\text{SO}_4) = \frac{c(\text{H}_2\text{SO}_4)_{\text{dil}} \cdot V(\text{H}_2\text{SO}_4)_{\text{dil}}}{V(\text{H}_2\text{SO}_4)} = \frac{0,01207 \cdot 2000}{5,000} = 4,828 \text{ mol} \cdot \text{L}^{-1}$$

17.6.

At the beginning, the flask only contains H_2SO_4 $0,01207 \text{ mol} \cdot \text{L}^{-1}$.



conc. inicial	0,01207	0,01207	0
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conc. final	$0,01207 - x$	$0,01207 + x$	x
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$$K_{a2} = \frac{[\text{H}_3\text{O}^+][\text{SO}_4^{2-}]}{[\text{HSO}_4^-]} \rightarrow 1,02 \cdot 10^{-2} = \frac{(0,01207 + x) \cdot x}{0,01207 - x} \rightarrow x^2 + 0,02227x - 1,23114 \cdot 10^{-4} = 0$$

$$\text{Solving} \rightarrow x = 0,004585$$

$$[\text{H}_3\text{O}^+] = 0,01207 + 0,004585 = 0,01666 \text{ mol} \cdot \text{L}^{-1} \rightarrow \text{pH} = 1,778$$

$$\text{V}(\text{NaOH}) = 0,000 \text{ mL} \quad \text{pH} = 1,78$$

In the semi – equivalence point all the acid is in form of NaHSO_4 .

$$c(\text{NaHSO}_4) = \frac{n(\text{H}_2\text{SO}_4)}{V(\text{D})} = \frac{0,01207 \cdot 10,00}{10,00 + 12,07} = 0,005469 \text{ mol} \cdot \text{L}^{-1}$$



conc. inicial	0,005469	0	0
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conc. final	$0,005469 - x$	x	x
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$$1,02 \cdot 10^{-2} = \frac{x^2}{0,005469 - x} \rightarrow x^2 + 0,0102x - 5,57838 \cdot 10^{-5} = 0$$

Solving $\rightarrow x_1 = 3,944 \cdot 10^{-3}$; $x_2 = -1,414 \cdot 10^{-2} \rightarrow \text{pH} = 2,404$

V(NaOH) = 12,07 mL pH = 2,40

In the equivalence point all the acid is in form of Na_2SO_4 which has basic hydrolysis.

$$c(\text{Na}_2\text{SO}_4) = \frac{n(\text{H}_2\text{SO}_4)}{V(\text{D})} = \frac{0,1207}{10,00 + 24,14} = 0,003535 \text{ mol} \cdot \text{L}^{-1}$$

$$K_h = \frac{[\text{OH}^-][\text{HSO}_4^-]}{[\text{SO}_4^{2-}]} = \frac{K_w}{K_{a2}} = \frac{1,00 \cdot 10^{-14}}{1,02 \cdot 10^{-2}} = 9,80 \cdot 10^{-13}$$

Because the K_h value and the concentration are small, it is not necessary to do the exact calculation.

V(NaOH) = 24,14 mL pH = 7,00 (The value obtained by exact calculation is 7,06)

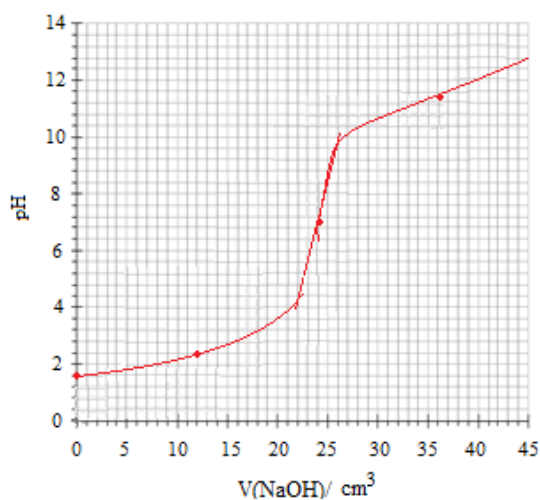
At the end, the flask contains Na_2SO_4 and NaOH in excess; sulfate hydrolysis can be ignored,

$$c(\text{NaOH}) = \frac{n(\text{NaOH})_{\text{exc}}}{V(\text{D})} = \frac{0,1207}{10,00 + 36,21}$$

$$c(\text{NaOH}) = 2,612 \cdot 10^{-3} \text{ mol} \cdot \text{L}^{-1} = [\text{OH}^-]$$

$$\text{pOH} = 2,583 \rightarrow \text{pH} = 11,42$$

V(NaOH) = 36,21 mL pH = 11,4

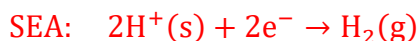
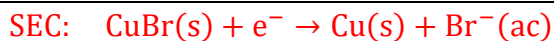


17.7.

bromothymol blue

Problem 18. Solubility of CuBr

18.1.



$$\Delta E = \Delta E^\circ - \frac{RT}{2F} \ln \left(\frac{c^2(\text{Br}^-)c^2(\text{H}^+)}{p(\text{H}_2)} \right)$$

18.2.

$$\Delta E^\circ = \Delta E + \frac{RT}{2F} \ln \left(\frac{c^2(\text{Br}^-)c^2(\text{H}^+)}{p(\text{H}_2)} \right) = 0,559 + \frac{8,314 \cdot 298}{2 \cdot 96486} \ln \left(\frac{(1,0 \cdot 10^{-4})^2 (1,0 \cdot 10^{-4})^2}{1,0} \right)$$

$$\Delta E^\circ = 0,086 \text{ V} = E^\circ(\text{CuBr/Cu})$$

18.3.

Using $\Delta G^\circ = -nF\Delta E^\circ$, one gets



$$K_{\text{ps}} = \exp \left(-\frac{\Delta G^\circ}{RT} \right) = \exp \left(-\frac{42100}{8,314 \cdot 298} \right) = 4,2 \cdot 10^{-8}$$

18.4.

$$[\text{Cu}^+] = \frac{K_{\text{ps}}}{[\text{Br}^-]} = \frac{4,2 \cdot 10^{-8}}{1,0 \cdot 10^{-4}} = 4,2 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

18.5.

Employing the Nernst equation

$$\Delta E(2) - \Delta E(1) = -\frac{RT}{2F} \ln \left(\frac{p(\text{H}_2)_1}{p(\text{H}_2)_2} \right) = -\frac{RT}{2F} \ln \frac{1}{2} = 0,0089 \text{ V}$$

Problem 19. Electrode potential chart

19.1.

Employing the formula

$$E^\circ = \frac{n_1 E^\circ_1 + n_2 E^\circ_2}{n_1 + n_2}$$

$$E^\circ(\text{Cr(V)}/\text{Cr(IV)}) = 1,35 \text{ V}; \quad E^\circ(\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}) = 1,33 \text{ V}; \quad E^\circ(\text{Cr}^{2+}/\text{Cr}) = -0,90 \text{ V}$$

19.2.

Species with tendency to disproportionate: Cr(V) and Cr(IV).



$$\Delta G^\circ_r = -zF\Delta E^\circ_r = -zF(E^\circ_{\text{red}} - E^\circ_{\text{ox}}) = -3 \cdot 96485 \cdot (-0,90 + 0,42) = 93 \text{ kJ} \cdot \text{mol}^{-1}$$

$$K = \exp\left(-\frac{\Delta G^\circ_r}{RT}\right) = \exp\left(-\frac{93000}{8,314 \cdot 298}\right) = 5,9 \cdot 10^{-17}$$

19.3.

The solubility constant can be calculated by comparing the values of the redox potentials corresponding to the electron pair Cr(III)/Cr at pH = 14 and at pH = 0.

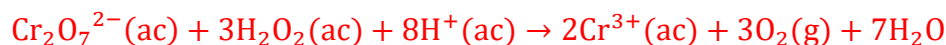
$$E = E^\circ - \frac{0,0592}{3} \log\left(\frac{1}{c(\text{Cr}^{3+})}\right) = E^\circ - \frac{0,0592}{3} \log\left(\frac{c^3(\text{OH}^-)}{K_{\text{ps}}}\right)$$

$$-1,33 - 0,74 - \frac{0,0592}{3} \log\left(\frac{1}{K_{\text{ps}}}\right) \rightarrow K_{\text{ps}} = 1,3 \cdot 10^{-30}$$

19.4.

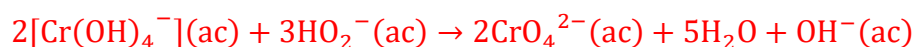
In acid conditions the largest possible potential difference is

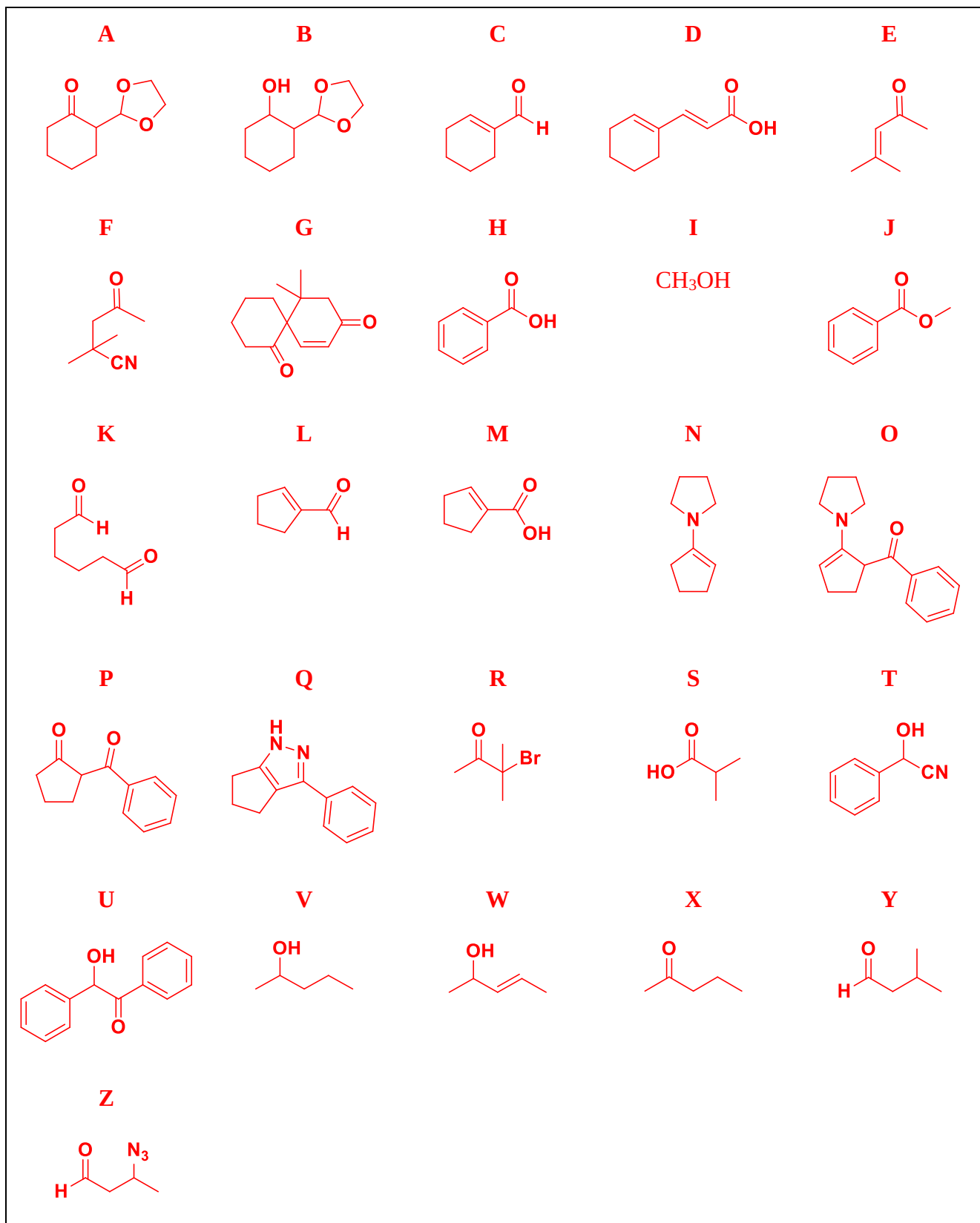
$$\Delta E^\circ_r = E^\circ(\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}) - E^\circ(\text{O}_2/\text{H}_2\text{O}_2) = 1,33 - 0,695 = 0,635 \text{ V}$$



In acid conditions the largest possible potential difference is

$$\Delta E^\circ_r = E^\circ(\text{HO}_2^-/\text{H}_2\text{O}) - E^\circ(\text{CrO}_4^{2-}/[\text{Cr}(\text{OH})_4^-]) = 0,87 + 0 + 0,72 = 1,59 \text{ V}$$



Problem 20. Aldehydes and ketones

Problem 21. Reaction mechanisms

21.1.

